Palo Alto Firewall
Initial Setup and Base Configuration

1. Objective

This lab provides a complete beginner guide to configuring a Palo Alto VM-Series Firewall from initial setup to full traffic validation including interfaces, zones, routing, NAT, security policies, and monitoring.

Test Environment Details

Virtualization Platform: VMware Workstation

Firewall Version: PAN-OS 11.2.0

Client Machine: Ubuntu 24.04 LTS

Source Zone: Trust

Destination Zone: Untrust

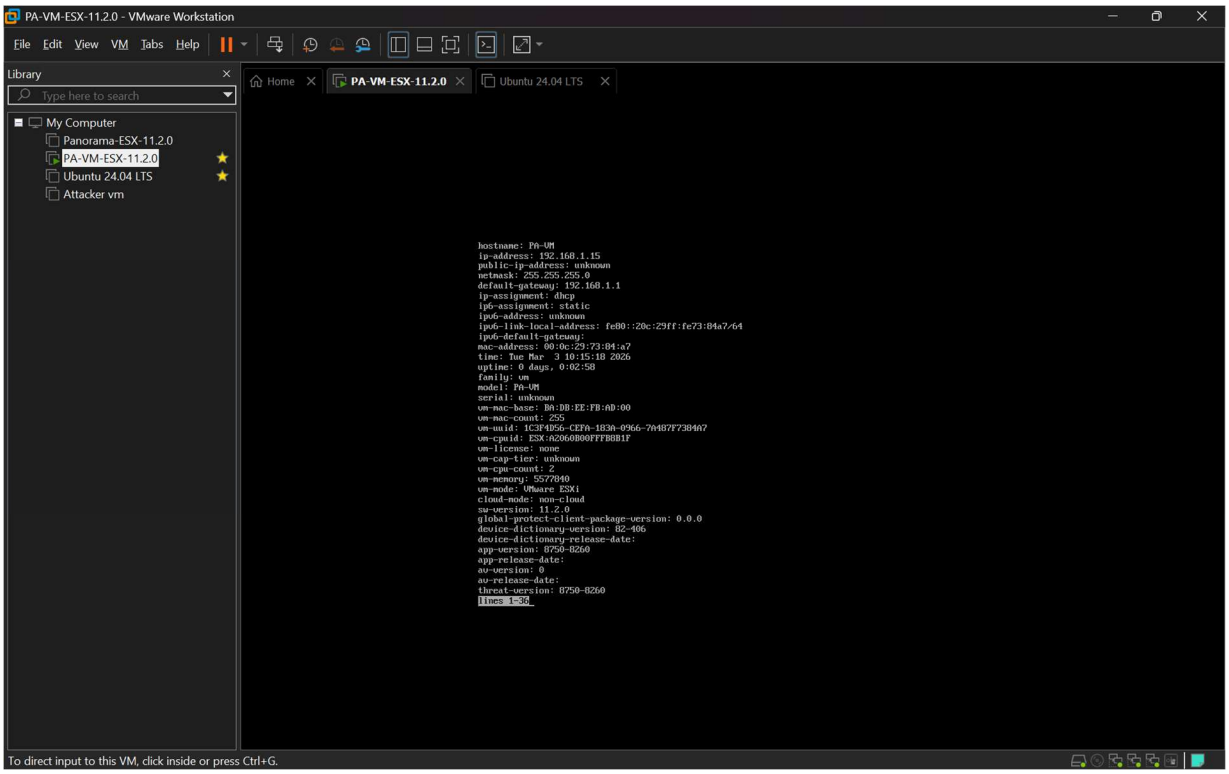
Network Topology Diagram

The below diagram represents the lab architecture where the Ubuntu VM connects to the internet through the Palo Alto Firewall.

2. Initial Firewall Configuration (First-Time Setup)

Step 1: Power on the Palo Alto VM.
Step 2: Login via console using admin/admin.
Step 3: Enter configuration mode using 'configure'.
Step 4: Configure management IP:
  set deviceconfig system ip-address 192.168.1.15
  set deviceconfig system netmask 255.255.255.0
  set deviceconfig system default-gateway 192.168.1.1
Step 5: Configure DNS server:
  set deviceconfig system dns-setting servers primary 8.8.8.8
Step 6: Commit configuration using 'commit'.
Step 7: Access Web UI via https://192.168.1.15

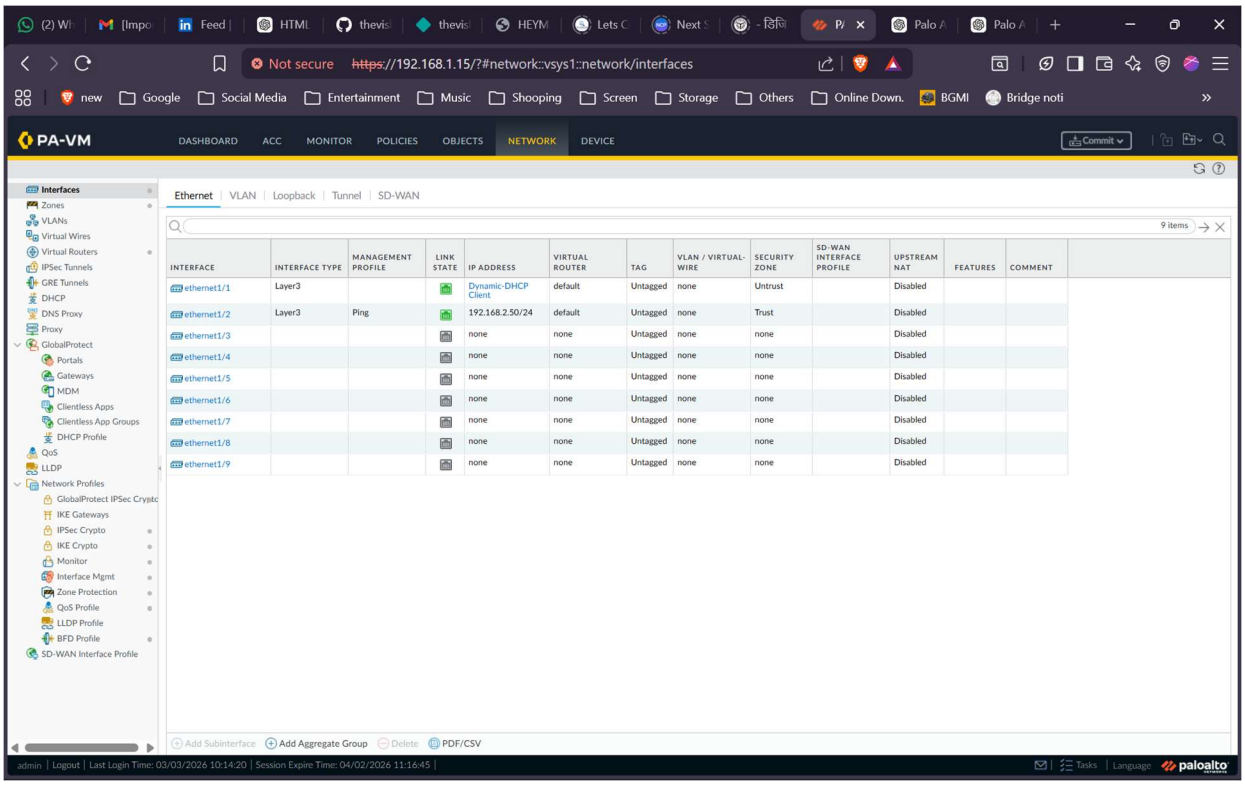
Result After Configuration:



3. Configure Layer 3 Interfaces (WAN and LAN Setup)

Navigate to Network > Interfaces.
Configure ethernet1/1 as Untrust (DHCP).
Configure ethernet1/2 as Trust (Static 192.168.2.50/24).
Assign to default virtual router.
Attach management profile (Ping enabled).
Commit changes.

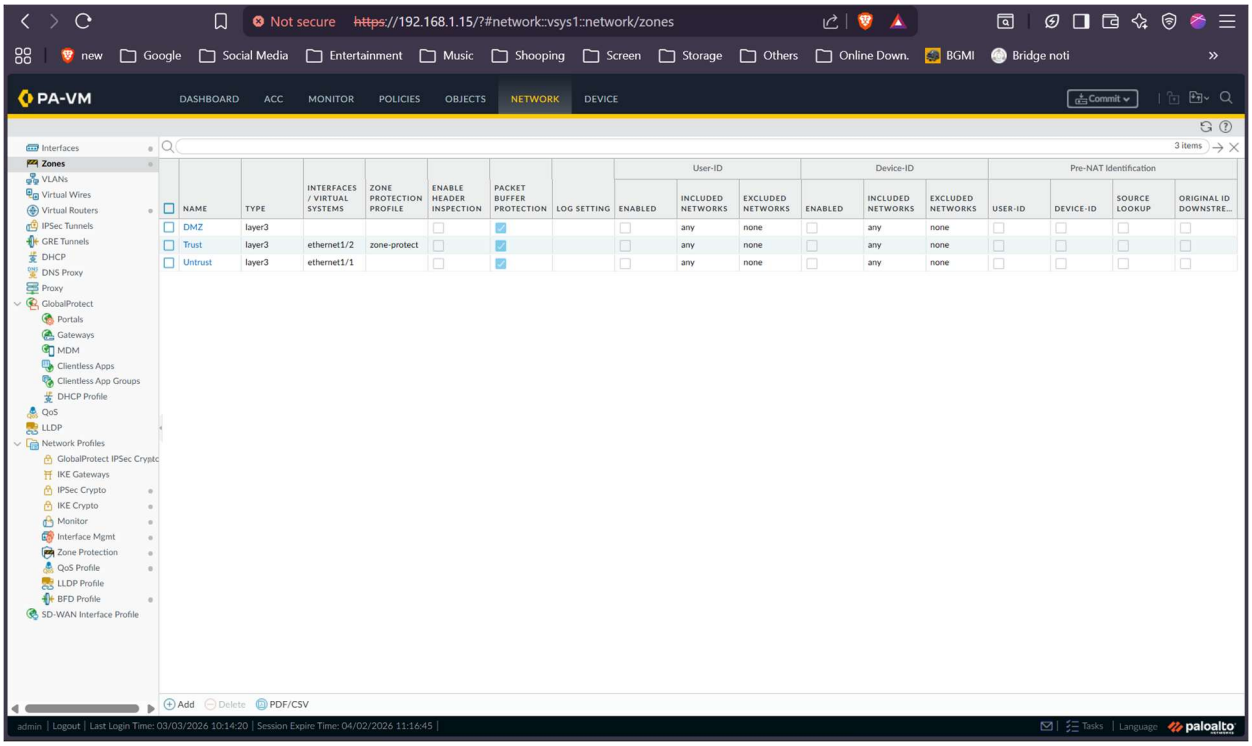
Result After Configuration:



4. Configure Security Zones

Navigate to Network > Zones.
Create Trust zone → assign ethernet1/2.
Create Untrust zone → assign ethernet1/1.
Commit configuration.

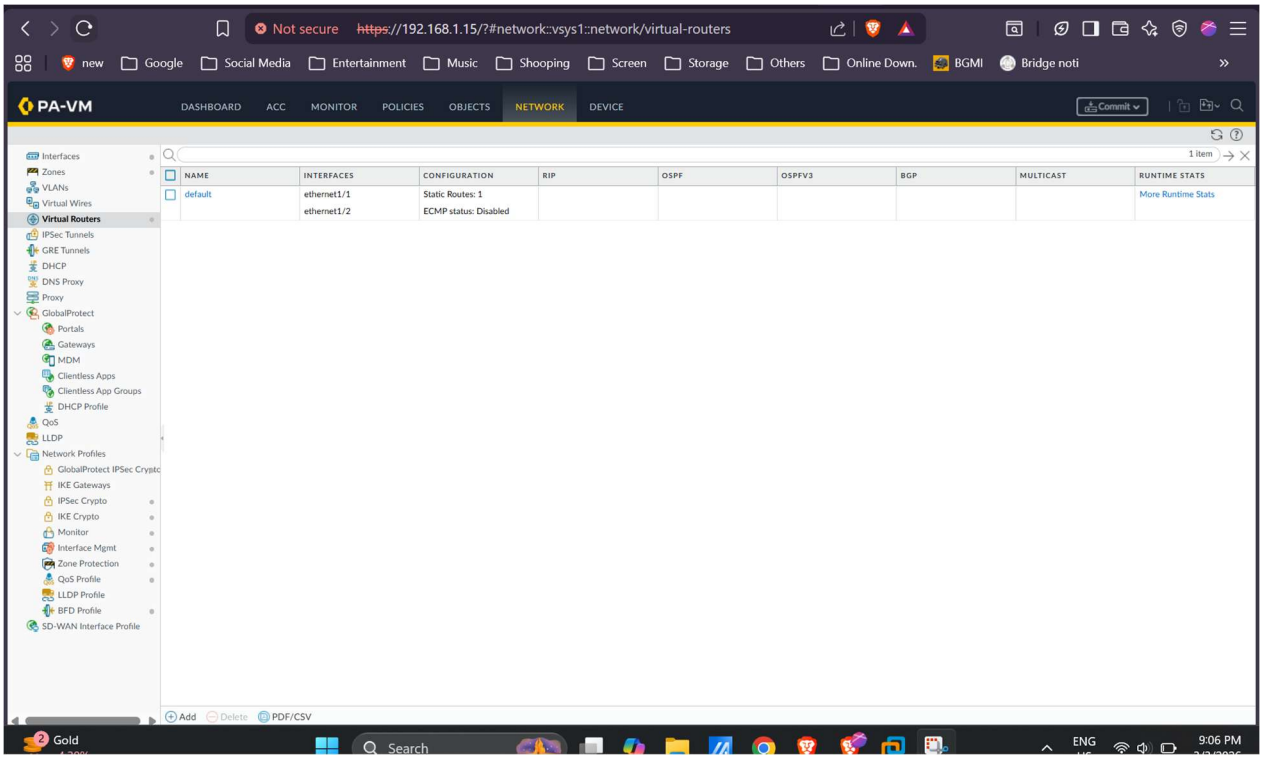
Result After Configuration:



5. Configure Virtual Router and Default Route

Navigate to Network > Virtual Routers > default.
Add static route:
Destination: 0.0.0.0/0
Next Hop: ISP gateway
Commit configuration.

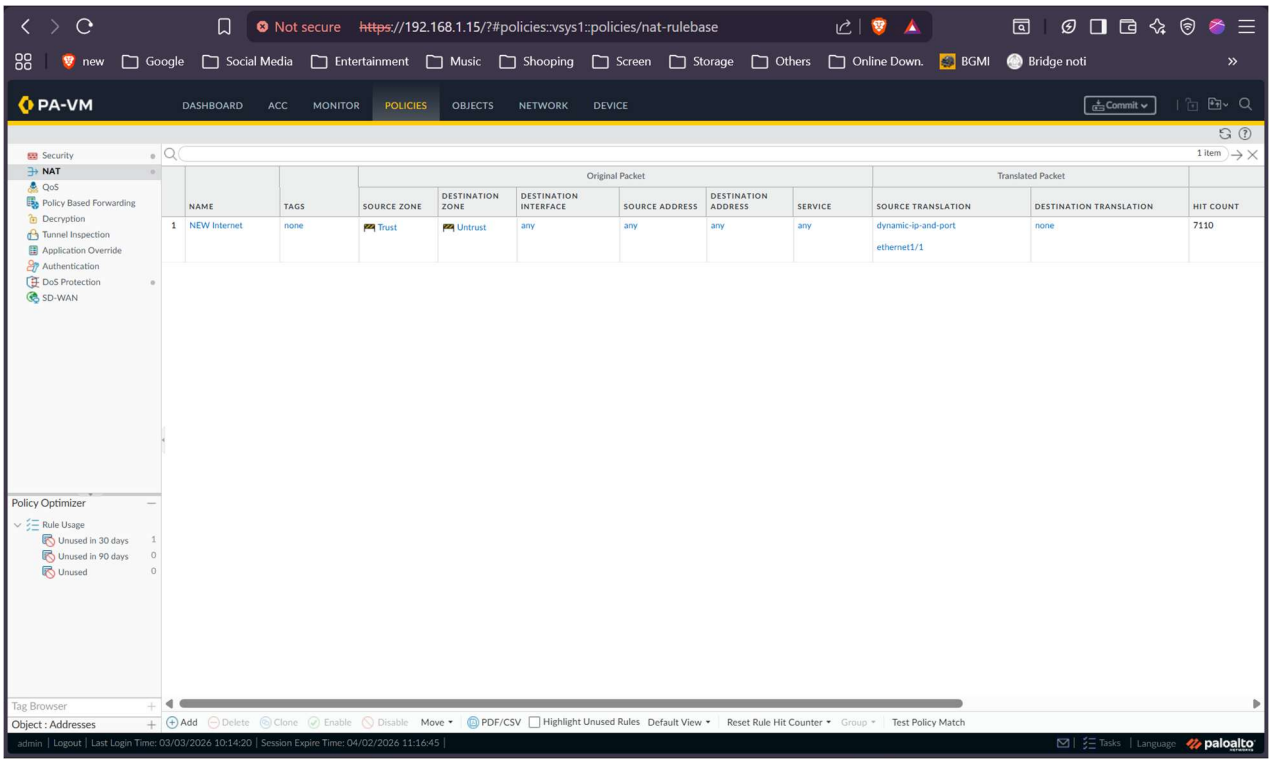
Result After Configuration:



6. Configure NAT Policy

Navigate to Policies > NAT.
Create rule 'NEW Internet'.
Source Zone: Trust
Destination Zone: Untrust
Source Translation: Dynamic IP and Port
Outgoing Interface: ethernet1/1
Commit configuration.

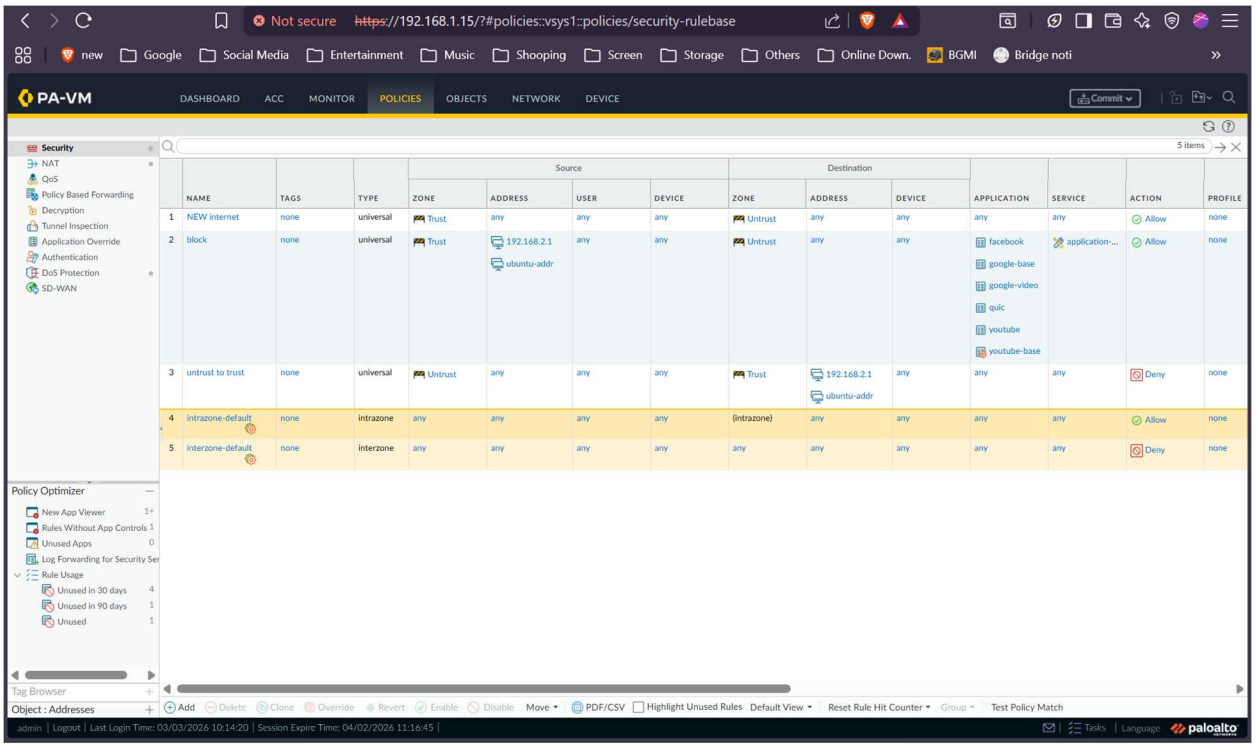
Result After Configuration:



7. Configure Security Policy

Navigate to Policies > Security.
Create allow rule from Trust → Untrust.
Application: any or selected apps.
Action: Allow.
Move rule above default rules.
Commit configuration.

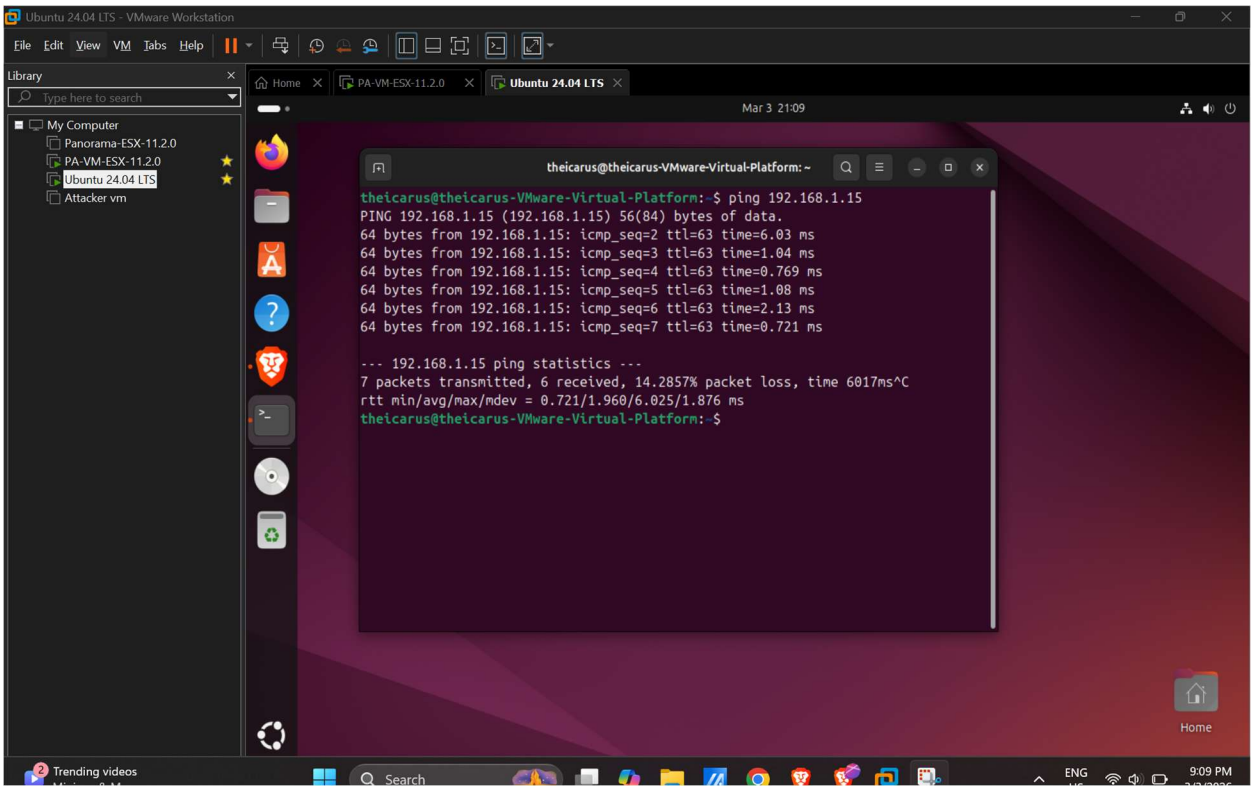
Result After Configuration:



8. Connectivity Verification (Ping Test)

From Ubuntu terminal run:
ping 192.168.1.15
Verify successful replies.

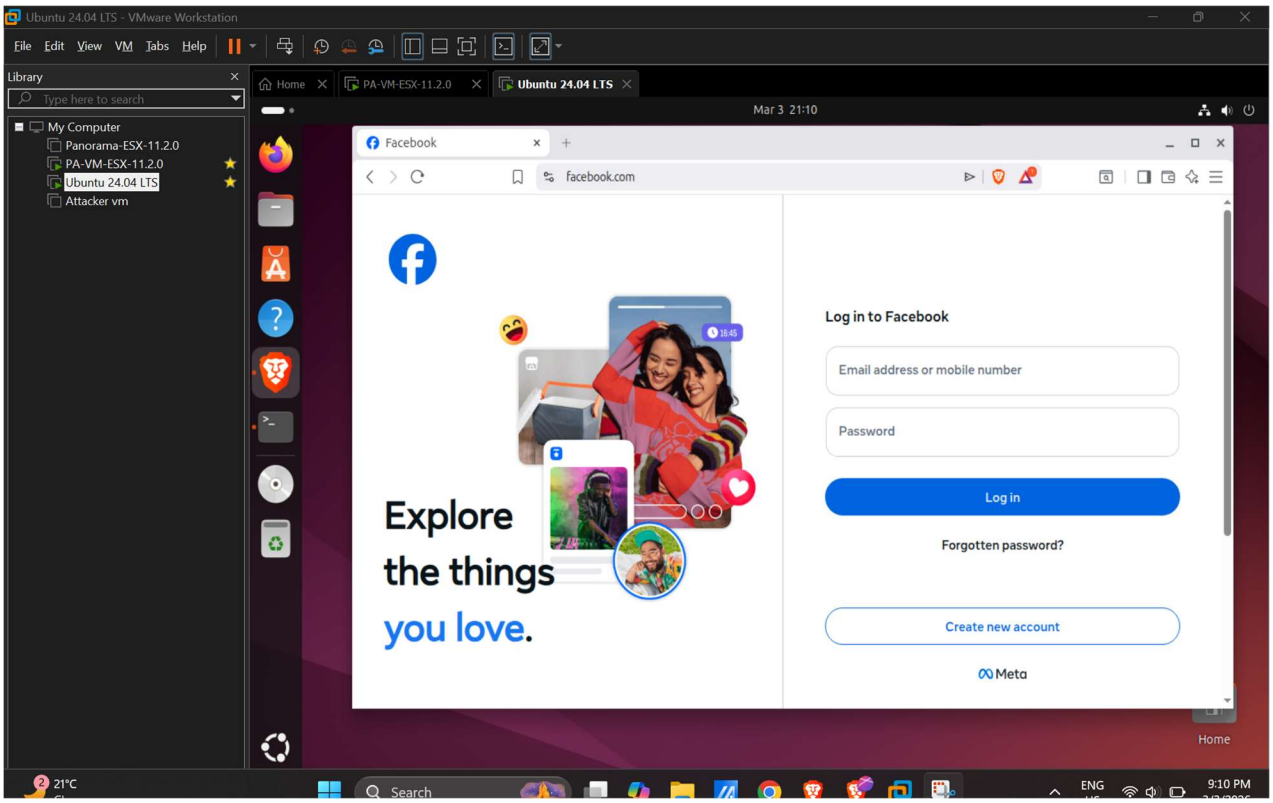
Result After Configuration:



9. Internet Access Verification

Open browser on Ubuntu.
Access https://facebook.com.
Verify page loads successfully.

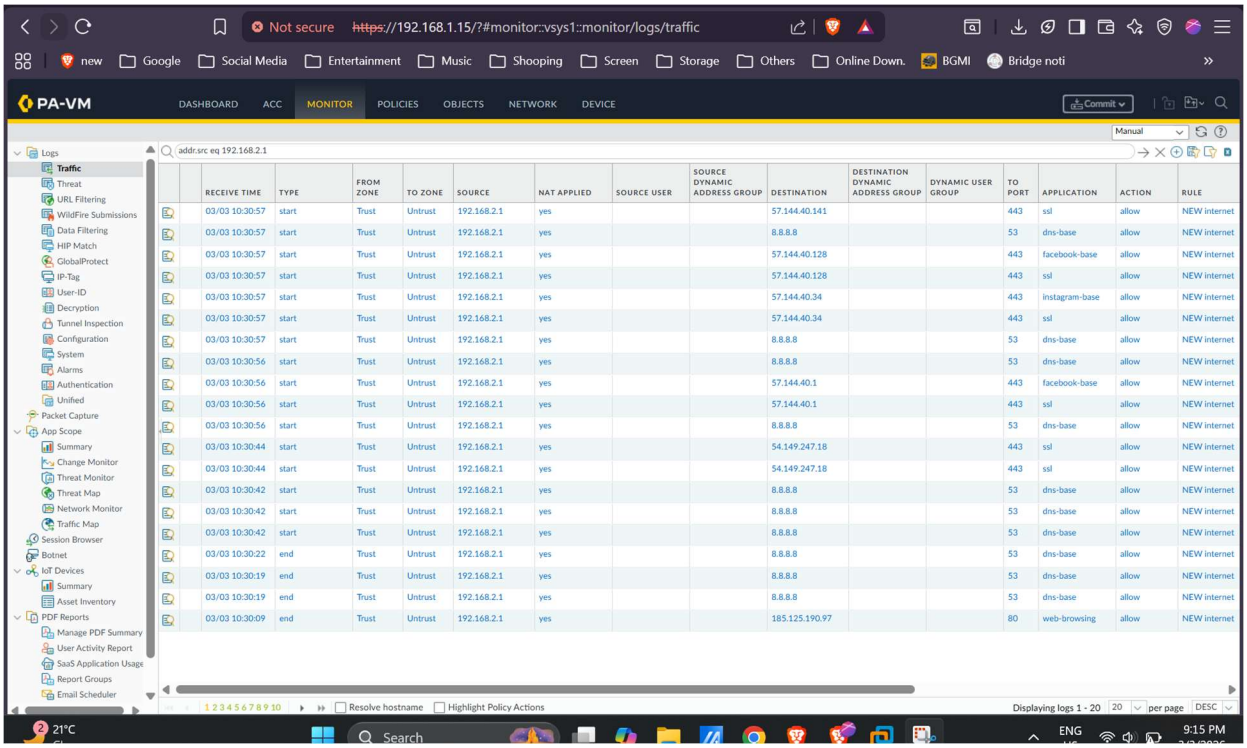
Result After Configuration:



10. Traffic Log Verification

Navigate to Monitor > Logs > Traffic.
Filter by source IP.
Verify NAT Applied = Yes.
Verify Application identified.
Verify Action = Allow.

Result After Configuration:



RECEIVE TIME	TYPE	FROM ZONE	TO ZONE	SOURCE	NAT APPLIED	SOURCE USER	SOURCE DYNAMIC ADDRESS GROUP	DESTINATION	DESTINATION DYNAMIC ADDRESS GROUP	DYNAMIC USER GROUP	TO PORT	APPLICATION	ACTION	RULE
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			57.144.40.141			443	ssl	allow	NEW internet
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			57.144.40.128			443	facebook-base	allow	NEW internet
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			57.144.40.128			443	ssl	allow	NEW internet
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			57.144.40.34			443	instagram-base	allow	NEW internet
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			57.144.40.34			443	ssl	allow	NEW internet
03/03 10:30:57	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:56	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:56	start	Trust	Untrust	192.168.2.1	yes			57.144.40.1			443	facebook-base	allow	NEW internet
03/03 10:30:56	start	Trust	Untrust	192.168.2.1	yes			57.144.40.1			443	ssl	allow	NEW internet
03/03 10:30:56	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:44	start	Trust	Untrust	192.168.2.1	yes			54.149.247.18			443	ssl	allow	NEW internet
03/03 10:30:44	start	Trust	Untrust	192.168.2.1	yes			54.149.247.18			443	ssl	allow	NEW internet
03/03 10:30:42	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:42	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:42	start	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:22	end	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:19	end	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:19	end	Trust	Untrust	192.168.2.1	yes			8.8.8.8			53	dns-base	allow	NEW internet
03/03 10:30:09	end	Trust	Untrust	192.168.2.1	yes			185.125.190.97			80	web-browsing	allow	NEW internet

11. Conclusion

The Palo Alto firewall was successfully configured from initial setup to full traffic validation. All components including interfaces, zones, routing, NAT, policies, and monitoring were implemented correctly.

- Vishesh Dixit